

Electrical Engineer Interim Report

Claim ID	PROP-B2B-2026-0417
Insured	Nordhafen Precision Parts GmbH
Document type	Electrical expert report
Issued by	ElektroSach GmbH
Document date	20 May 2026
Loss location	Hafenstrasse 18, Building B, 30179 Hannover, Germany

Assignment and evidence reviewed

ElektroSach GmbH was instructed to conduct an interim electrical origin-and-cause review. The assignment was limited to non-destructive examination, photographs, breaker panel inspection, and visual assessment of recovered components. Laboratory testing had not yet been performed.

Evidence item	Observed condition
48V charger HL-48V-70A	Severe exterior thermal damage; internal transformer area heavily charred; serial plate partly legible.
Blue multi-socket extension lead	Socket body melted; copper conductors exposed; no industrial rating label visible.
Wall socket B-LB-17	Faceplate distorted; copper terminals showed discoloration; breaker B16 tripped.
Pallet truck battery	Outer case heat damaged on charger side; no obvious cell rupture in visual inspection.
Cable tray above alcove	Soot and heat plume marks rising from charger bench area.

Preliminary findings

- The most probable area of origin is within 1.5 metres of the charging bench, not the adjacent maintenance room.
- Damage pattern is consistent with an electrical fault at or near the charger supply connection, but laboratory confirmation is required.
- The extension lead appears unsuitable for continuous overnight industrial charging loads. Its recovered position indicates it may have supplied the charger.
- Wall socket and breaker condition could be cause or effect; sequence cannot yet be determined from visual inspection alone.
- No evidence of deliberate ignition was observed during the limited examination.

Conflict with service record

HanseLift Service GmbH record HL-2026-0504 states that the charger cable was replaced and load-tested on 04 May 2026. The recovered charger cable shows older hardening and discoloration near the plug. Either the service record does not refer to the recovered cable, or the cable was exposed to abnormal heating after service.

Electrical distribution observations

Circuit / component	Finding	Comment
Panel LB-2 / breaker B16	Tripped; labelled "Packing bench sockets" not "charger"	Labelling inconsistent with fixed charging-station warranty.
RCD group 2	Tripped at arrival, reset not attempted	No insulation resistance test before reset.
Wall socket B-LB-17	16A Schuko outlet; not interlocked industrial socket	May be ordinary socket circuit.
Charger isolation switch	Not found	No dedicated isolator visible within 1 metre of charger.
Nearby combustibles	Cardboard sleeves and polymer inserts stored approx. 0.8 metre away	Contrary to two-metre separation if confirmed.

Limitations

This interim report does not determine final cause. It does not exclude battery internal failure, defective charger manufacture, poor service work, overloaded socket circuit, or external fire spread. Laboratory examination and interviews with service technicians are recommended.

Recommended next steps

- Secure full service file from HanseLift, including photographs, technician notes, and parts invoice.
- Obtain original electrical installation drawings and commissioning certificate for charging area.
- Send charger, plug, wall socket, breaker, and extension lead to laboratory for metallurgical and arc mapping analysis.
- Compare CCTV timestamps with alarm log and fire brigade timeline.